A Role-Resolved Source-Function Audit of the White et al. Sphere–Cylinder Casimir Geometry

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Abstract

White et al.'s sphere-in-cylinder Casimir result contains two separable parts: a boundary-geometry observation and a warp-source interpretation. The boundary-geometry question concerns shape: whether a $1\,\mu$m diameter sphere centered in a nominal $4\,\mu$m diameter cylinder organizes the vacuum-boundary proxy into the annular region where the reported warp-like shell appears. The source-function question concerns role and scale: whether that organized region remains a shell-placed source proxy under channel accounting, how its calibrated magnitude compares with an Alcubierre source demand, and how the central readout path participates.

This paper evaluates the claim ladder with two role-resolved proxy layers. A pair-resolved v-loop scalar morphology probe supplies the shape read: each dimensionless contribution is a loop-center/scale event whose segments cross both the sphere surface and the inner cylinder-wall surface, followed by the declared scale weight. A finite-difference tensor-scale audit then supplies the source-function read: common random loops, parallel-plate Casimir calibration brackets, source-demand comparison, linearized metric bounds, material-energy estimates, and explicit readout accounting. The sphere, cylinder wall, annular shell, central readout path, boundary infrastructure, ordinary electromagnetic competitor region, and far-field control are scored as separate roles.

The integrated result gives White's geometry a clear physical interpretation. The selective scalar channel concentrates signed proxy magnitude in the annular source region, places the radial maximum near the sphere-wall gap, and keeps far-field leakage small. The tensor-scale proxy preserves this shell placement and assigns a small fraction of the source proxy to the central readout role. This supports the shell-morphology interpretation and clarifies the readout path as a measurement role distinct from the main source channel. SI calibration then fixes the magnitude layer: the source-demand ratio is of order 10^{-39} in the most favorable tested row, and the linearized timing bound is of order 10^{-76} s. The geometry therefore reads as a real shell-shaped Casimir boundary morphology with Casimir-scale gravitational strength, giving useful structure to the Alcubierre analogy while locating warp-functionality at a source scale many orders above the achieved value.

1. QUESTION

White et al. applied worldline numerics to custom Casimir geometries and found a negative energy-density morphology that they described as intersecting with the energy-density requirements of the Alcubierre metric [1]. The most compact version of that reported geometry is the sphere-in-cylinder toy model: a $1\,\mu$m diameter sphere centered in a $4\,\mu$m diameter cylinder. The reported calculation produced a shell-like negative distribution that was visually close to a two-dimensional section of an Alcubierre-style negative-energy shell.

This study evaluates the claim ladder behind that comparison. The first layer is scalar morphology: whether the sphere-cylinder boundary geometry organizes signed pair occupancy into the proposed shell under scale, seed, wall, and geometry variations. Here "robust" means concentration in the intended sphere-cylinder shell region under baseline seed changes, wall-thickness changes, reported scale windows, and single-seed modest sphere and cylinder perturbations.

The second layer is source-function relevance. A scalar shell silhouette and an Alcubierre source occupy different claim layers. The stronger question asks where the role-accounted morphology sits under finite-difference source proxies, how the calibrated magnitude compares with a representative Alcubierre source demand, how the central readout path participates, and how material and ordinary electromagnetic competitors enter a laboratory interpretation. The evidence is therefore organized as morphology, role placement, scale closure, material competition, and readout accounting.

The analysis protocol draws on prior source-placement and role-separation work [5, 6]. The White device is treated here in its own Casimir terms: roles, channels, controls, and observables are separated before source or transport language is assigned.

2. GEOMETRY AND ROLES

The evaluated geometry is the White et al. sphere-cylinder toy model:

$$d_s = 1.0\,\mu\text{m}, \qquad d_c = 4.0\,\mu\text{m}, \tag{1}$$

where d_s and d_c are nominal diameters and the sphere is centered in the cylinder. The crossed cylinder surface is the inner wall surface

$$R_{\text{inner}} = R_c - \frac{t_{\text{wall}}}{2}. \tag{2}$$

For the $d_c = 4.0\,\mu$m baseline, the reported $0.2\,\mu$m and $0.4\,\mu$m wall rows use inner crossing diameters of $3.8\,\mu$m and $3.6\,\mu$m. The analysis treats the cylinder wall as boundary infrastructure, the sphere as the stress-shaping body, the annular region outside the sphere and inside the cylinder as the source-shell candidate, the central axis as the proposed transit/readout channel, and the exterior region as far-field control.

For a grid point $x = (x, y, z)$ in a two-dimensional section, let $r_s(x)$ be the distance from the sphere center and let $r_\perp(x) = \sqrt{y^2 + z^2}$ be the transverse radius relative to the cylinder axis. The near-shell band used for the principal concentration metric is

$$\mathcal{S}_{\text{shell}} = \{x :\ R_s < r_s(x) \le R_s + 0.75\,\mu\text{m}, \quad r_\perp(x) < R_c - 0.10\,\mu\text{m}\}. \tag{3}$$

This fixed nominal band is a predefined role-labeled accounting region used for all reported cases to represent the intended sphere-cylinder shell. The material wall, central readout mask, and far field are tracked with their own accounting masks, and those masks may overlap on grid cells where geometric roles share space.

3. LOOP CONSTRUCTION AND KERNEL

The loop generator follows the v-loop construction quoted in White et al. The sampled Gaussian variables use the paper's $\exp(-w_i^2)$ convention, i.e. a normal distribution with variance 1/2. Let $y_{a,n} \in \mathbb{R}^3$ denote point n on loop a, with the loop center of mass subtracted.

The scalar proxy uses segment/surface crossings. For each translated and scaled loop

$$\Gamma_{a,\lambda,c} = \{c + \lambda y_{a,n}\}_{n=1}^{N}, \tag{4}$$

the kernel determines whether any segment of the loop crosses the sphere surface and whether any segment crosses the inner cylinder wall. A loop contributes to the pair interaction when both crossings occur:

$$I_{a,\lambda}(c) = \mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial B_s \neq \varnothing\}\mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial C_{\text{inner}} \neq \varnothing\}. \tag{5}$$

The normalized scalar morphology proxy at grid center c is then

$$\mathcal{E}_{\mathrm{proxy}}(c) = -\frac{1}{N_{\mathrm{loop}}} \sum_{\lambda \in \Lambda} \frac{1}{\lambda^4} \sum_{a=1}^{N_{\mathrm{loop}}} I_{a,\lambda}(c). \tag{6}$$

The λ^{-4} weighting keeps the scalar proxy aligned with the expected scale sensitivity of a four-dimensional Casimir-type contribution. The sum over λ is a declared discrete geometry-scale score for comparing morphology within the same support interval. A stress-tensor or energy-density estimator adds continuum proper-time quadrature, quadrature weights, point-splitting or derivative operators, material/boundary response, and SI normalization. The output category here is a normalized scalar morphology proxy with explicit dimensionless occupancy status.

The single-body controls are embedded in the scoring rule. Contributions are accepted when a loop crosses both the sphere and the inner cylinder surface. This makes the probe a direct pair-interaction read; generic boundary contacts carry zero contribution in this score.

4. READOUT METRICS

Let

$$M_-(c) = \max[-\mathcal{E}_{\mathrm{proxy}}(c), 0] \tag{7}$$

be the signed-occupancy magnitude recorded by the proxy. The principal shell concentration metric is

$$F_{\mathcal{S}_{\mathrm{shell}}} = \frac{\sum_{c \in \mathcal{S}_{\mathrm{shell}}} M_-(c)}{\sum_c M_-(c)}. \tag{8}$$

The shell enrichment is the ratio between mean signed magnitude in the near-shell band and mean signed magnitude outside the band:

$$Q_{\mathcal{S}_{\mathrm{shell}}} = \frac{\langle M_- \rangle_{\mathcal{S}_{\mathrm{shell}}}}{\langle M_- \rangle_{\mathbb{R}^2 \setminus \mathcal{S}_{\mathrm{shell}}}}. \tag{9}$$

The top-decile shell fraction records the fraction of the strongest signed pixels that fall in $\mathcal{S}_{\mathrm{shell}}$. Boundary, far-field, and readout participation are computed as the corresponding magnitude fractions in their role-labeled regions. Two central-channel quantities are tracked: the explicit readout path from the geometry role label, with radius $0.05\,\mu$m, and a wider central-axis mask,

$$A_{0.15} = \{x : \ r_\perp(x) \le 0.15\,\mu\mathrm{m}\}. \tag{10}$$

The explicit readout path is an accounting mask applied after the scalar field is computed. The crossing geometry keeps the sphere and cylinder boundary model fixed; a modeled bore, conductor, material removal, or propagation observable is a separate readout calculation. The wider $A_{0.15}$ mask records nearby central-axis participation on the grid.

5. COMPUTATION CONFIGURATION

The calculation used the configuration in Table 1. The computational design emphasizes the interaction kernel and role-accounted scale windows. Loop count and grid size set the scope of the present scalar morphology calculation. A normalized White-style scalar comparison adds a larger loop ensemble, continuum proper-time scaling, parallel-plate normalization, and direct comparison to the reported 2000-loop run. The source-function layer below adds a local finite-difference tensor-scale proxy, SI calibration brackets, source-demand comparison, material estimates, and readout accounting. A full worldline stress tensor and a modeled laboratory propagation observable belong to a higher-fidelity physical layer. The validation scope for this manuscript is crossing geometry, sign accounting, role bookkeeping, scale-window morphology, source-role stability, and calibrated scale comparison.

Table 1: Pair-surface scalar-probe configuration.

Item	Value
Kernel	Pair-resolved segment/surface crossing
Surfaces	Sphere surface and inner cylinder wall
Loop method	Paper-style v-loop
Baseline seeds	$1, 7, 17$
Wall thicknesses	$0.2\,\mu$m, $0.4\,\mu$m
Inner crossing diameters for $d_c = 4.0\,\mu$m	$3.8\,\mu$m, $3.6\,\mu$m
Baseline scale windows	0.75–$1.50\,\mu$m, 1.50–$2.75\,\mu$m, 0.25–$5.00\,\mu$m
Geometry perturbations	sphere $0.9\,\mu$m, sphere $1.1\,\mu$m, cylinder $3.8\,\mu$m, cylinder $4.2\,\mu$m
Perturbation seed	1
Loops	96
Points per loop	192
Grid	37×37 over $\pm 2.5\,\mu$m
Workers	4

6. BASELINE RESULTS

Table 2 gives the baseline aggregate over the three seeds. The mid-scale window is the most spatially selective morphology channel. It covers about 0.35–0.37 of the sampled pixels with signed occupancy values, yet it places 0.647–0.726 of the signed magnitude in the fixed nominal near-shell band. Boundary and far-field fractions are negligible, and the peak radial bin remains at $1.125\,\mu$m.

Table 2: Baseline pair-surface scalar-probe read. Values are means over seeds $1, 7, 17$.

Window	Wall	Neg. frac.	$F_{\mathcal{S}_{\text{shell}}}$	$Q_{\mathcal{S}_{\text{shell}}}$	Top-decile shell	Boundary	Far
0.75–1.50	0.2	0.347699	0.646612	9.782	0.781228	0.000283	0.000000
0.75–1.50	0.4	0.365961	0.726420	14.206	0.883074	0.001319	0.000000
1.50–2.75	0.2	0.892866	0.540825	6.289	0.771857	0.012854	0.005076
1.50–2.75	0.4	0.893109	0.567666	7.012	0.875146	0.023294	0.000718
0.25–5.00	0.2	1.000000	0.504602	5.437	0.708029	0.020623	0.014162
0.25–5.00	0.4	1.000000	0.547679	6.464	0.790754	0.032290	0.002884

The upper-scale window is broader and fills most of the section, and its largest signed pixels remain shell-concentrated. The broad scale-sum window gives apparatus-wide signed scalar participation because larger loops connect the sphere-wall system across much of the sampled section. Even in the broad field, approximately half of the signed magnitude remains in the fixed nominal near-shell band with 5–6 fold enrichment. These rows separate two scalar channels: apparatus-wide broad-window participation and localized mid-scale pair interaction.

Table 3 gives the corresponding central-channel read. In the mid-scale channel, the explicit readout path carries 0.000651–0.001225 of the signed proxy magnitude, and the wider central-axis mask carries 0.002918–0.005459. The upper-scale and broad channels add apparatus-wide central participation at the few-percent level on the explicit readout path.

Table 3: Baseline readout-axis participation. Values are means over seeds $1, 7, 17$.

Window	Wall	Explicit readout path	Central axis $A_{0.15}$
0.75–1.50	0.2	0.000651	0.002918
0.75–1.50	0.4	0.001225	0.005459
1.50–2.75	0.2	0.019870	0.063433
1.50–2.75	0.4	0.023333	0.073679
0.25–5.00	0.2	0.023119	0.071958
0.25–5.00	0.4	0.023688	0.073302

7. GEOMETRY PERTURBATIONS

Table 4 summarizes the mid-scale perturbation cases at seed 1. In this single-seed geometry-direction check, the tighter cylinder strengthens the near-shell concentration, the wider cylinder weakens it, and both sphere-size perturbations retain a recognizable shell-localized response. The peak radial bin remains near $1.125\,\mu$m in the mid-scale cases.

Table 4: Mid-scale perturbation read at seed 1.

Geometry	Wall	Neg. frac.	$F_{\mathcal{S}_{\text{shell}}}$	$Q_{\mathcal{S}_{\text{shell}}}$	Top-decile shell
Cylinder $3.8\,\mu$m	0.2	0.379839	0.718409	13.618	0.865385
Cylinder $3.8\,\mu$m	0.4	0.385683	0.782528	19.208	0.962264
Cylinder $4.2\,\mu$m	0.2	0.310446	0.539340	6.250	0.627907
Cylinder $4.2\,\mu$m	0.4	0.372535	0.632558	9.189	0.745098
Sphere $0.9\,\mu$m	0.2	0.344777	0.628413	9.658	0.708333
Sphere $0.9\,\mu$m	0.4	0.352082	0.719429	14.643	0.897959
Sphere $1.1\,\mu$m	0.2	0.395179	0.689049	10.860	0.818182
Sphere $1.1\,\mu$m	0.4	0.401753	0.762371	15.723	0.909091

The upper-scale perturbation cases retain a broader and still organized read: near-shell magnitude fractions range from 0.515 to 0.595, top-decile near-shell fractions range from 0.719 to 0.942, and far-field magnitude fractions stay below 0.006655. The perturbation rows give the same geometry-direction picture as the baseline calculation: the mid-scale channel is localized, and larger scale channels add apparatus-wide participation while the largest magnitudes remain near the shell. Across the perturbation cases, the mid-scale explicit readout-path fraction stays in the 0.000000–0.002634 range, with the wider central-axis fraction in the 0.000000–0.008408 range. The upper-scale perturbation cases carry 0.017245–0.027398 on the explicit readout path and 0.055520–0.084184 on the wider central-axis mask.

8. TENSOR-SCALE SOURCE-FUNCTION PROBE

The source-function layer keeps the same role separation and asks a stronger question: whether the scalar morphology remains source-shell placed under paired finite-difference source proxies, whether a calibrated magnitude is competitive with an Alcubierre source demand, and whether the central readout path carries the source proxy. The calculation is a finite-difference source-channel proxy. Its role is to bind the morphology result to transparent scale and channel accounting before a full worldline stress tensor is attempted.

The focused tensor-scale calculation used a 73×73 grid over $\pm 2.5\,\mu$m, 12 loop blocks, 32 loops per block, 256 points per loop, the two selective scale windows 0.75–1.50 μm and 1.50–2.75 μm, wall thicknesses 0.2 μm and 0.4 μm, a 0.025 μm finite-difference step, and the same 0.05 μm readout accounting radius. It produced 48 tensor blocks, 4032 stress-channel rows, 1536 source-demand rows, and 576 material-competitor rows. The calibration stage used parallel-plate Casimir brackets for ideal EM and scalar Dirichlet families over gaps 0.75–2.75 μm; gap ordering passed in all calibration families and scale windows.

Table 5 gives the baseline role fractions for the negative-magnitude source proxy. The mid-scale channel is almost entirely a source-shell channel. The upper-scale channel is broader, with larger stress-body and ordinary electromagnetic competitor participation, and it remains shell dominated.

Table 5: Tensor-scale role fractions for the finite-difference source proxy.

Window	Wall	Shell	Readout	EM region	Body	Far
0.75–1.50	0.2	0.995759	0.000231	0.000679	0.004074	0.000000
0.75–1.50	0.4	0.992679	0.000340	0.001962	0.007203	0.000000
1.50–2.75	0.2	0.938713	0.009849	0.019400	0.050144	0.003941
1.50–2.75	0.4	0.932350	0.011070	0.031461	0.057991	0.000807

The finite-difference channel read identifies the geometry controls that move the proxy. The source-shell channel responds primarily to cylinder inner radius and sphere diameter. In the mid-scale rows, the cylinder-radius shell medians are 541.8 and 739.5 per micrometer for wall thicknesses 0.2 μm and 0.4 μm, while the sphere-diameter shell medians are -345.0 and -492.1 per micrometer. In the upper-scale rows, the corresponding shell medians are 366.5, 386.9, -314.1, and -340.6 per micrometer. Cylinder length is essentially null at this scale, and perturbing the readout accounting radius acts on the readout channel without generating a source-shell response. That pattern anchors the shell result to the sphere-cylinder boundary geometry and keeps readout-mask bookkeeping in its own accounting role.

Table 6 summarizes the calibrated scale comparison. The best source-demand ratio occurs in the ideal-EM mid-scale family at material factor 1.0, target width 2.0 μm, and speed parameter 10^{-9}, and is 1.019×10^{-39}. The largest linearized timing bound is 7.81×10^{-76} s. These numbers are scale bounds. A transport prediction requires a propagation model, while the present source-demand table places the shell-like morphology many orders below Alcubierre source-demand magnitude in this harness.

Table 6: Calibrated source and metric-scale bounds.

Quantity	Value
Maximum source-demand ratio	1.019×10^{-39}
Full source-demand ratio range	3.20×10^{-62}–1.019×10^{-39}
Integrated shell energy range	5.24×10^{-25}–2.35×10^{-21} J
Readout-coupled energy range	0–1.42×10^{-23} J
Maximum gravitational radius	2.34×10^{-67} m
Maximum fractional path scale	2.93×10^{-62}
Maximum timing bound	7.81×10^{-76} s

The readout summary distinguishes source-placement accounting from a later laboratory propagation measurement. Source placement asks how much of the proxy is assigned to the central path; propagation asks how photons, electrons, or current respond to the surrounding material geometry. This source-function probe directly tests source placement. Propagation observables require their own optical, electronic, or current-transport calculation. In the calibrated summary, the mid-scale readout source fraction is zero; in the upper-scale summary, the median readout source fraction is 0.010449 while the shell channel fraction is 0.934467, a shell/readout ratio of about 89.4. The source proxy is therefore shell-first, with readout retained as a distinct measurement role.

The material and ordinary electromagnetic estimates set the laboratory interpretation boundary. Across the material table, the Casimir-to-material rest-energy ratio ranges from 1.06×10^{-29} to 3.95×10^{-25}, and the patch-potential-to-Casimir energy ratio ranges from 2.57 to 1.15×10^{6}. Patch-potential and contact-loading flags appear in every material row. These estimates leave the shell morphology intact and set a strong condition on experimental interpretation: ordinary material and electromagnetic effects require explicit modeling.

9. INTERPRETATION

Within these role-resolved proxy layers, the White et al. sphere-cylinder geometry produces a near-shell Casimir boundary morphology. The mid-scale scalar channel gives the most spatially discriminating read: signed proxy magnitude concentrates in the annular source band, far-field magnitude is small, boundary participation remains secondary, and the radial peak stays near the middle of the sphere-wall gap. The tensor-scale source proxy preserves the same source-shell placement and keeps the central readout channel small.

The broad-window field describes a separate scale channel. Larger loops connect the apparatus over most of the sampled section, producing apparatus-wide signed scalar participation. The broad field retains a shell-enriched largest-magnitude component and combines localized sphere-wall morphology with long-scale apparatus connection. The Alcubierre-like comparison is carried by the separated mid-scale pair interaction, which localizes around the intended shell band.

The readout-axis metrics describe overlap between the proxy fields and central accounting masks. In the clean mid-scale channel, most signed proxy magnitude is assigned to the near-shell band. The explicit transit/readout mask receives about one tenth of one percent of the scalar magnitude and zero calibrated tensor-scale readout source fraction. Larger scale channels add weak central participation as part of the broader apparatus field. The source proxy is concentrated where a shell would be drawn, while the central path appears mainly through apparatus-scale accounting.

The source-demand comparison supplies the scale-limiting result. The Alcubierre comparison requires stress components, sign placement, magnitude comparison, and metric source demand in addition to scalar morphology. In the present calibrated proxy, the morphology is strong enough to be a real shell-placement result, and the source magnitude sits many orders beneath a representative Alcubierre demand. Recent work by Garattini and Zatrimaylov frames this same layer in terms of energy conditions and metric-source structure for warp-drive geometries [4]. A proposed transit-time or current/photon/electron readout also carries ordinary electromagnetic competition: capacitance, impedance, dispersion, material response, contact loading, patch potentials, surface roughness, and geometry-dependent leakage can all dominate a laboratory signal.

10. CODE AND DATA AVAILABILITY

Code and generated data for this study are available in the GitHub repository `https://github.com/dgoldman0/spacetime-metric-engineering`. The relevant package is `white_casimir_source_function_audit` in the repository's `white_casimir_intersection/experiments` tree. That package contains the scalar fidelity case table, aggregate summary, mean field arrays, tensor-scale harness, finite-difference stress-channel proxy, Alcubierre source-demand comparison, linearized metric-bound estimate, ordinary-EM proxy, and pytest checks for the local crossing geometry, scoring, bookkeeping, and Stage 3 harness.

The focused tensor-scale output used for the source-function tables is stored under the external research run directory:

```
/media/kir/9CDCBD3EDCBD140C/Research/
white_casimir_tensor_scale_probe/stage3_focused_20260530
```

The compact output files are:

- `calibration/calibration_summary.csv`
- `tensor/stress_channel_summary.csv`
- `tensor/bootstrap_summary.csv`
- `scale/source_demand_ratio.csv`
- `scale/linearized_metric_bound.csv`
- `material/material_competitor_summary.csv`
- `readout/readout_coupling_summary.csv`

11. CONCLUSION

The audit gives White et al.'s sphere-cylinder observation a sharper physical identity. The apparatus behaves, in these role-resolved proxies, as a structured Casimir boundary system. The selective scalar field and tensor-scale source proxy both place the strongest accounting in the annular sphere-wall band, with far-field leakage small and the readout path retained as a minor measurement channel. The Alcubierre resemblance therefore has a concrete basis at the morphology/source-placement layer: the geometry can shape vacuum-boundary proxy structure into the same kind of shell role drawn in the metric comparison.

That result is useful because it separates shape, source placement, gravitational scale, and laboratory readout into distinct physical claims. The shape claim receives the strongest support. The source-placement claim also receives support inside the finite-difference proxy: the source proxy follows the shell channel across the selective windows, and the bore path contributes a small counted share. The metric claim receives a scale assignment. The calibrated proxy places the gravitational/source magnitude many orders beneath representative Alcubierre source demand. Refinements may move the calibration bracket; the present scale gap is the controlling physical fact.

This scale assignment changes the narrative from a visual coincidence into a quantified boundary-physics result. White's geometry is valuable as a reproducible shell-forming Casimir configuration and as a test case for source-placement auditing. Its warp relevance, in this audit class, is geometric and organizational: it tells us where the source-like proxy lives, how sharply it stays in the shell role, and how small the associated metric scale is after SI calibration. That combination is scientifically meaningful even at Casimir gravitational strength, because it gives future calculations a well-defined target and removes ambiguity about which part of the apparatus is carrying the shell analogy.

The remaining experimental question is a propagation and materials question. A central-channel timing, photon, electron, or current readout depends on finite conductivity, patch potentials, contact loading, waveguide behavior, surface roughness, detector coupling, and the chosen material stack. Those effects enter at the same scale as the proposed observable and require a dedicated transport calculation tied to the actual apparatus. The present result therefore preserves the interesting physics in White et al.'s observation, gives the Alcubierre comparison its proper role, and sets the next calculation on firmer ground: a shell-shaped Casimir boundary morphology with calibrated source scale, material competition, and explicit readout propagation handled as coupled parts of a single physical model.

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